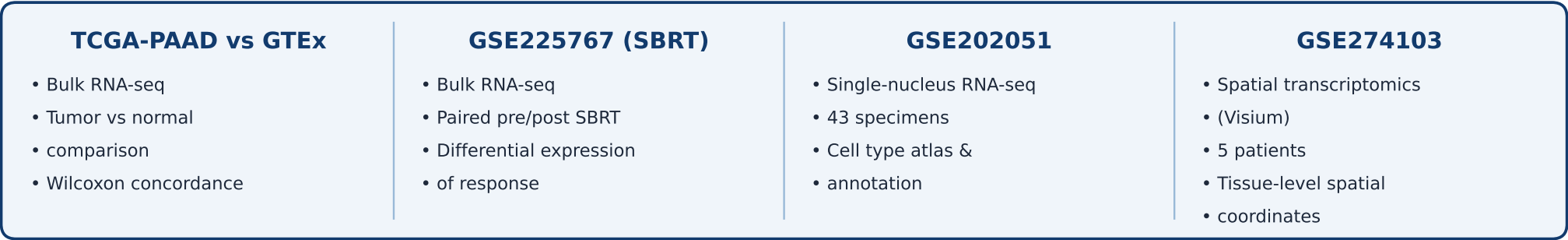
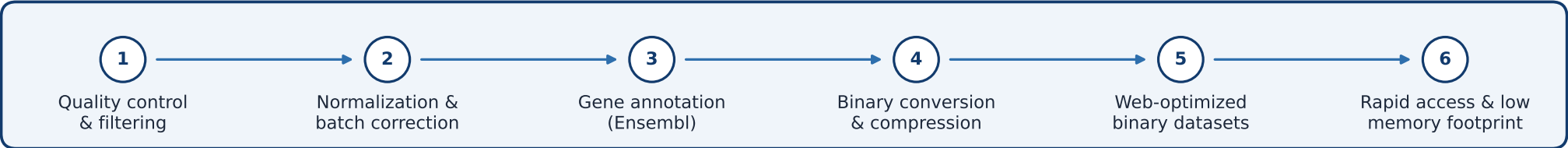


PDAC-BioPortal: System Architecture

1. PUBLIC TRANSCRIPTOMIC DATASETS

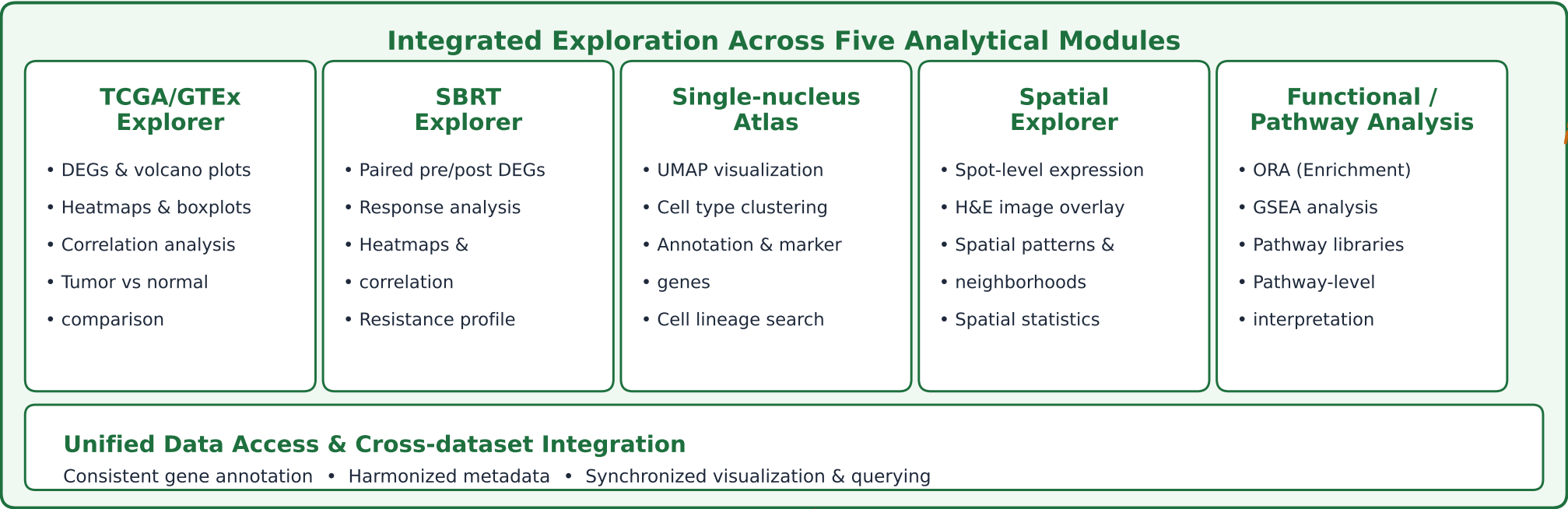


2. DATA PROCESSING & STORAGE

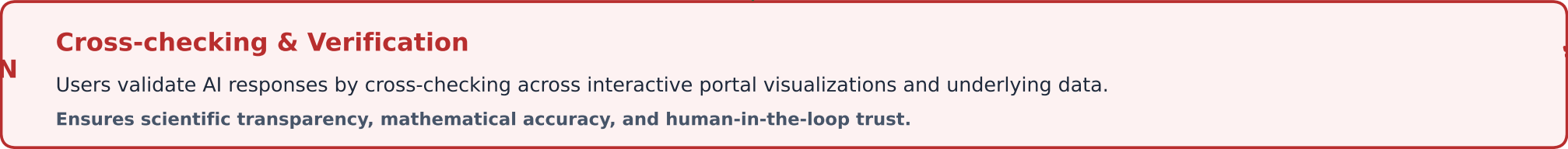


3. PDAC-BioPortal (Next.js Web Interface)

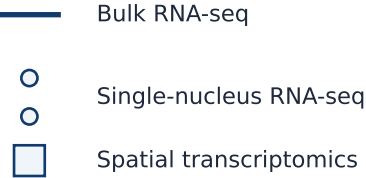
Core Analytical Platform



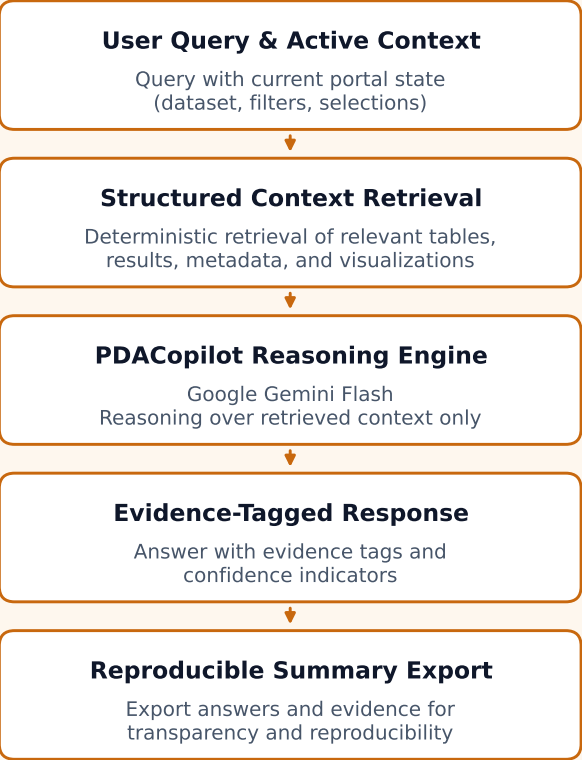
5. USER VERIFICATION (Human-in-the-Loop)



DATA TYPES



PDACopilot AI Assistant (Decoupled Layer)



DESIGN PRINCIPLES

- ✓ Modular & scalable
- ✓ Data integrity & reproducibility
- ✓ Transparent & evidence-driven
- ✓ Decoupled AI for reliability

